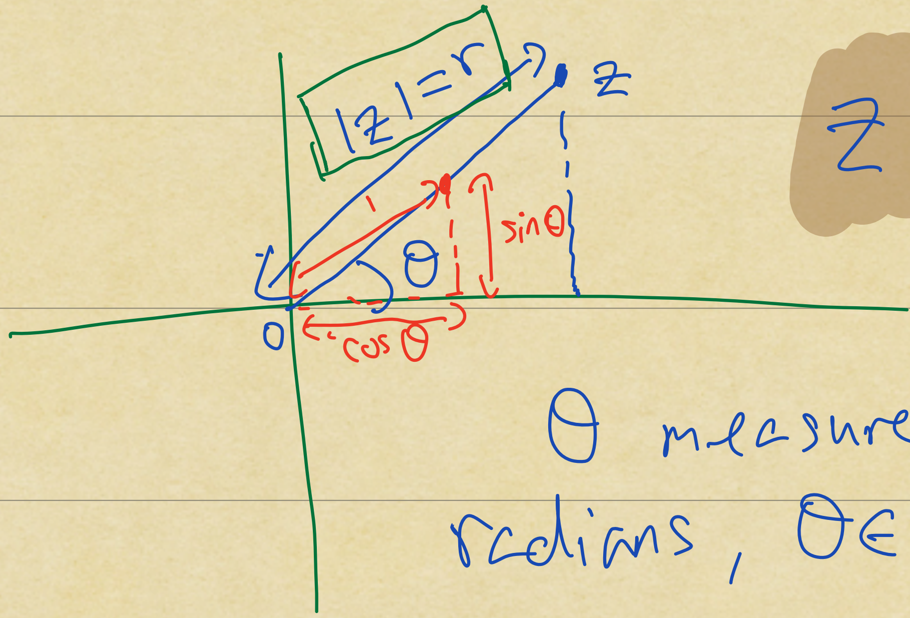
Complex numbers

Euler's notation



$$z = re^{i\theta}$$
$$= re^{\theta i}$$

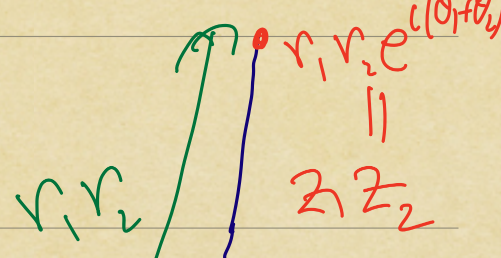
θ measured in radians, $\theta \in [0, 2\pi)$

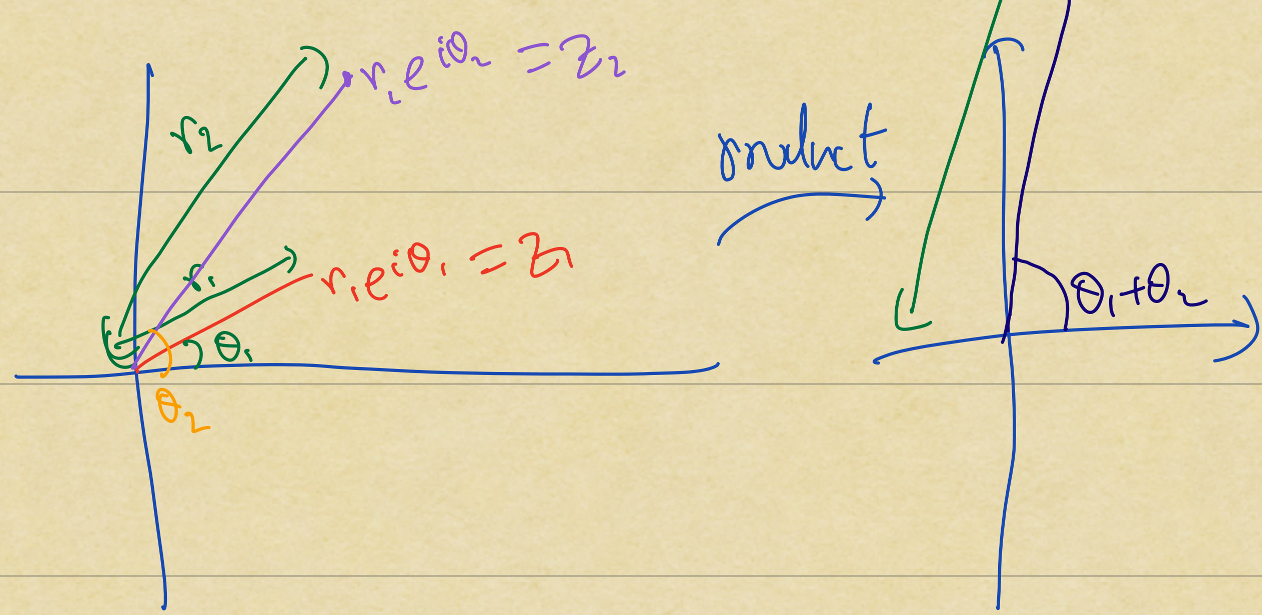
Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Trig formulas $\Longleftrightarrow$

$$e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$$





Example:

Solve $z^3 = 1$ in $\mathbb{C}$

Write $z = \boxed{r} e^{\boxed{i\theta}}$

Then we need to solve

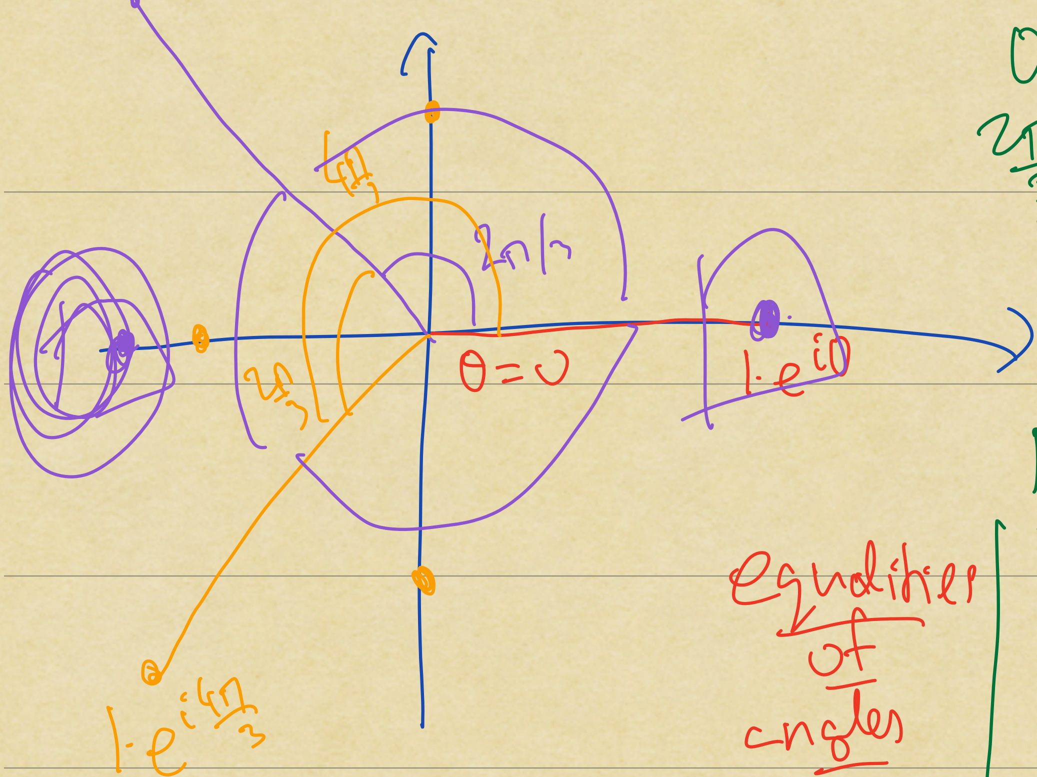
$$(re^{i\theta})^3 = 1$$

$$\Updownarrow$$

$$\boxed{r^3 e^{i3\theta}} = r^3 (e^{i\theta})^3 = \boxed{1}$$

$$\Rightarrow \quad r^3 = 1 \Leftrightarrow r = 1$$

$1 \cdot e^{i\frac{2\pi}{3}}$



$0\cdot 3 = 0$

$\frac{2\pi}{3}\cdot 3 = 2\pi \underset{\text{angle}}{\overset{\text{as}}{=}} 0$

$\frac{4\pi}{3}\cdot 3 = 4\pi \underset{\text{angle}}{\overset{\text{as}}{=}} 0$

Note:

equalities of angles

$$\boxed{\begin{aligned} \frac{6\pi}{3} &= 2\pi = 0 \\ \frac{8\pi}{3} &= \frac{2\pi}{3} \end{aligned}}$$

Rmk: the same technique applies for the equation

$$z^n = \alpha$$

for fixed $\alpha \in \mathbb{C}$
$\alpha \neq 0$

There are n solutions

If $\alpha = s e^{i\theta}$

then the solutions are

$$z = s^{1/n} e^{i(\theta + 2\pi k)/n}$$

$$k = 0, 1, \ldots, n-1$$

Takeaways

Examples of fields

Finite	Infinite
$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$
p prime	rational, real, complex

Thm (Galois)

The only possible sizes of finite fields are powers of primes p^d. Moreover, for every such power, there is a unique field of size p^d.

Example:

F: field

$a, b, c \in F$; then we can write down a

polynomial $\boxed{a}x^2 + \boxed{b}x + \boxed{c} = f(x)$

Can we find $f(x)$ s.t.

$f(1) = 1 \iff a \cdot 1^2 + b \cdot 1 + c = 1$

$f(-1) = -1 \iff a \cdot (-1)^2 + b \cdot (-1) + c = -1$

$f(0) = 1 + 1 = 2 \neq 0$?

$\iff a \cdot 0 + b \cdot 0 + c = 2$

a, b, c are our variables for a linear system

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\downarrow R2 - R1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\downarrow R2 \times (-2)^{-1}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

This would fail if $-2 = 0 \in F$

$\downarrow R1 - R2$

e.g. if
F=4/24

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$\downarrow R1 - R3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

Solution

$$f(x) = -2x^2 + x + 2$$

Rmk: What is happening here is that the space of polynomials of deg ≤ 2 is a vector space over F, and we are solving a linear problem in this vector space

Vector space

F : field

Def$\underline{^n}$ A vector space V over F is a set V equipped with:

(zero) (a) $0 \in V$ [also $0_V \in V$]

(Addition) (b) $+ : V \times V \to V$

$(v_1, v_2) \mapsto v_1 + v_2$

(c) (Scalar multiplication) $\cdot : \underbrace{\boxed{F}}^{\text{Scalars}} \times V \to V$

$(a, v) \mapsto a \cdot v$

Satisfying:

① (Additive identity) $0 + v = v + 0 = v \quad \forall v \in V$

② (Additive inverse) $\forall v \in V, \exists -v \in V$ s.t. $v + (-v) = 0$

③ (Associativity) $v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3 \quad \forall v_1, v_2, v_3 \in V$

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(Commutativity)

$v_1 + v_2 = v_2 + v_1$

$\forall v_1, v_2 \in V$

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(Distributivity)

$a \cdot (v_1 + v_2)$

$= a \cdot v_1 + a \cdot v_2$

$\forall a \in F$

$v_1, v_2 \in V$

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(Associativity for scalar multiplication)

$a_1 \cdot (a_2 \cdot v) = (a_1 a_2) \cdot v$

$\forall a_1, a_2 \in F$

$v \in V$